pharmaniaga®

Metronidazole

Tablet

DESCRIPTION
Pharmaniaga Metronidazole tablet 200 mg
Yellow, round, biconvex, film-coated tablets with 'RMB' breakline on one side and plain on the other.

Pharmaniaga Metronidazole tablet 400 mg
Yellow, round, biconvex tablet with Pharmaniaga icon on one side and plain on the other.

COMPOSITION
Pharmaniaga Metronidazole tablet 200 mg
Each film-coated tablet contains Metronidazole 200 mg.

Pharmaniaga Metronidazole tablet 400 mg
Each tablet contains Metronidazole 400 mg.

PHARMACODYNAMICS
Metronidazole is active against most obligate anaerobic bacteria and protozoa by undergoing intracellular chemical reduction via mechanisms unique to anaerobic metabolism within the organism. Reduced metronidazole, which is cytotoxic but short-lived, interacts with microorganisms DNA to cause a loss of helical structure, strand breakage and resultant inhibition of nucleic acid synthesis and cell death.

PHARMACOKINETICS
Metronidazole is rapidly absorbed from the gastro-intestinal tract. Peak plasma concentrations occur between 20 minutes and 3 hours. The elimination half-life is 8.5±2.9 hours. Metronidazole can be used in chronic renal failure; it is rapidly removed from the plasma by dialysis. Metronidazole is excreted in milk but the intake of a suckling infant of a mother receiving normal dosage would be considerably less than the therapeutic dosage for infants.

INDICATIONS
Metronidazole is indicated for the treatment of symptomatic trichomoniasis in females and males when the presence of trichomonad has been confirmed by appropriate laboratory procedures (wet smears and/or cultures).

Metronidazole is indicated in the treatment of asymptomatic females when the organism is associated with endocervicitis, cervicitis or cervical erosion. Since there is evidence that presence of the tricomonad can interfere with accurate assessment of abnormal cytological smears, additional smears should be performed after eradication of the parasite.

T. vaginalis infections is a venereal disease. Therefore, asymptomatic sexual partners of treated patients should be treated, simultaneously if the organism has been found to be present, in order to prevent reinfection of the partner. The decision as to whether to treat an asymptomatic male-partner who has negative culture or one for whom no culture has been attempted is an individual one. In making this decision, it should be noted that there is evidence that a woman may become reinfected if her consort is not treated. Also, since there can be considerable difficulty in isolating the organism from the asymptomatic male carrier, negative smears and cultures cannot be relied upon in this regard. In any event, the consort should be treated with metronidazole in cases of reinfection.

Metronidazole is indicated in the treatment of acute intestinal amoebiasis (amoebic dysentery) and amoebic liver abscess.

In amoebic liver abscess, metronidazole therapy does not obviate the need for aspiration or drainage of pus. Metronidazole is indicated in the treatment of serious infections caused by susceptible anaerobic bacteria. Indicated surgical procedures should be performed in conjunction with metronidazole therapy. In a mixed aerobic and anaerobic infection, antimicrobials appropriate for the treatment of the aerobic infection should be used in addition to metronidazole.

In the treatment of most serious anaerobic infections, intravenous metronidazole is usually administered initially. This may be followed by oral therapy at the discretion of the physician.

Intra-abdominal infections, including peritonitis, intra-abdominal abscess, and liver abscess caused by *Bacteroides* species including the *B. fragilis* group *(B. fragilis, B. distasonis, B. ovatus, B. thetaiotaomicron, B. vulgatus), Clostridium* species, *Eubacterium* species, *Peptococcus niger* or *Peptostreptococcus* species.

Skin and skin structure infections caused by *Bacteroides* species including the *B. fragilis* group. *Clostridium* species, *Peptococcus niger, Peptostreptococcus* species or *Fusobacterium* species.

Gynaecologic infections including endometritis, endomyometritis, tubo-ovarian abscess, and postsurgical vaginal cuff infection, caused by *Bacteroides* species including the *B. fragilis* group. *Clostridium* species, *Peptococcus niger* or *Peptostreptococcus* species.

Bacterial septicaemia caused by *Bacteroides* species including the *B. fragilis* group or *Clostridium* species.

Bone and joint infections (as adjunctive therapy) caused by *Bacteroides* species including the *B.fragilis* group.

Central nervous system (CNS) infections, including meningitis and brain abscess, caused by *Bacteroides* species including the *B.fragilis* group.

Endocarditis caused by *Bacteroides* species including the *B. fragilis* group.

CONTRAINDICATIONS
Metronidazole is contraindicated in patients with a prior history of hypersensitivity to metronidazole or other nitroimidazole derivatives; in patients with trichomoniasis, during the first trimester of pregnancy.

ADVERSE REACTIONS
The following reactions have also been reported during treatment with metronidazole.

Central Nervous System: The most serious adverse reactions reported in patients treated with metronidazole have been convulsive seizures, encephalopathy, aseptic meningitis, optic and peripheral neuropathy, the latter characterised mainly by numbness or paraesthesia of an extremity. Since persistent peripheral neuropathy has been reported in some patients receiving prolonged administration of metronidazole, patients should be told to stop the drug and report immediately to their physicians if any neurologic symptoms occur. In addition, patients have reported headache, syncope, dizziness, drowsiness, vertigo, incoordination, ataxia, confusion, convulsion, dysarthria, irritability, depression, weakness, insomnia, gait impairment, nystagmus and tremor.

Gastro-intestinal: The most common adverse reactions reported have been referable to the gastro-intestinal tract, particularly nausea reported by about 12% of patients, sometimes accompanied by headache, anorexia and occasionally vomiting, diarrhoea, epigastric distress and abdominal cramping. Constipation has also been reported.

A sharp, unpleasant metallic taste is not unusual. Taste disorder, oral mucositis, furred tongue, glossitis and stomatitis have occurred; these may be associated with sudden overgrowth of Candida which may occur during therapy. Rare cases of pancreatitis, which generally abated on withdrawal of the drug, have been reported.

Haematopoietic: Reversible neutropenia (leucopenia); rarely, reversible thrombocytopenia.

Cardiovascular: Flattening of the T-wave may be seen in electrocardiographic tracings.

Hypersensitivity: Urticaria, erythematous rash, Steven-Johnson syndrome, toxic epidermal necrolysis, pruritus, flushing, nasal congestion, dryness of the mouth (or vagina or vulva), fever, angioedema and anaphylaxis.

Skin and subcutaneous tissue: Skin rashes, erythematous rash, pustular eruptions, acute generalised exathematous pustulosis, pruritus, flushing, erythema multiforme, Steven-Johnson syndrome, toxic epidermal necrolysis and fixed drug eruption.

Blood and lymphatic system: Agranulocytosis, neutropenia, thrombocytopenia, pancytopenia and leucopenia.

Metabolism and nutrition: Anorexia.

Psychiatric: Psychotic disorders, including confusion and hallucinations and depressed mood.

Eye: Vision disorders such as diplopia and myopia (which, in most cases, is transient) and optic neuropathy/neuritis.

Ear and labyrinth: Hearing impaired/ hearing loss (including sensorineural) and tinnitus.

Hepatobiliary: Increase in liver enzymes (AST, ALT, alkaline phosphatase), cholestatic or mixed hepatitis and hepatocellular liver injury, jaundice and pancreatitis which is reversible on drug withdrawal. Cases of liver failure requiring liver transplant have been reported in patients treated with metronidazole in combination with other antibiotic drugs.

Musculoskeletal, connective tissue and bone: Myalgia and arthralgia.

Renal: Dysuria, cystitis, polyuria, incontinence and a sense of pelvic pressure. Instances of darkened urine have been reported by approximately one patient in 100,000. Although the pigment which is probably responsible for this phenomenon has not been positively identified, it is almost certainly a metabolite of metronidazole and seems to have no clinical significance.

Other: Proliferation of Candida in the vagina, dyspareunia, decrease of libido, proctitis and fleeting joint pains sometimes resembling "serum sickness". If patients receiving metronidazole drink alcoholic beverages, they may experience abdominal distress, nausea, vomiting, flushing or headache. A modification of the taste of alcoholic beverages has also been reported.

Patients with Crohn's disease are known to have an increased incidence of gastro-intestinal and certain extra-intestinal cancers. There have been some reports in the medical literature of breast and colon cancer in Crohn's disease patients who have been treated with metronidazole at high doses for extended periods of time. A cause and effect relationship has not been established. Crohn's disease is not an approved indication for metronidazole.

WARNINGS AND PRECAUTIONS
Cases of severe hepatotoxicity/ acute hepatic failure, including cases with a fatal outcome with very rapid onset after treatment initiation in patients with Cockayne syndrome have been reported with products containing metronidazole for systemic use. In this population, metronidazole should therefore be used after careful benefit-risk assessment and only if no alternative treatment is available. Liver function test must be performed just prior to the start of therapy, throughout and after end of treatment until liver function is within normal ranges, or until the baseline values are reached. If the liver function test become markedly elevated during treatment, the drug should be discontinued. Patients with Cockayne syndrome should be advised to immediately report any symptoms of potential liver injury to their physician and stop taking metronidazole.

Cases of encephalopathy and peripheral neuropathy (including optic neuropathy) have been reported with metronidazole. Encephalopathy has been reported in association with cerebellar toxicity characterized by ataxia, dizziness and dysarthria. CNS lesions seen on MRI have been described in reports of encephalopathy. CNS symptoms are generally reversible within days to weeks upon discontinuation of metronidazole. CNS lesions seen on MRI have also been described as reversible.

Peripheral neuropathy, mainly of sensory type has been reported and is characterized by numbness or paresthesia of an extremity. Convulsive seizures have been reported in patients treated with metronidazole.

Cases of aseptic meningitis have been reported with metronidazole. Symptoms can occur within hours of dose administration and generally resolve after metronidazole therapy is discontinued.

The appearance of abnormal neurologic signs and symptoms demands the prompt evaluation of the benefit/risk ratio of the continuation of therapy. Metronidazole should be administered with caution to patient with central nervous system disease.

Patients with severe hepatic disease metabolize metronidazole slowly, with resultant accumulation of metronidazole and its metabolites in the plasma and may contribute to the symptoms of the encephalopathy. This medication should therefore, be administered with caution to patients with hepatic encephalopathy. For patients with severe hepatic impairment (Child-Pugh C), a reduced dose of metronidazole is recommended (Refer Section Dosage and Administration). For patients with mild to moderate hepatic impairment, no dosage adjustment is needed but these patients should be monitored for metronidazole associated adverse events.

Known or previously unrecognised candidiasis may present more prominent symptoms during therapy with metronidazole and requires treatment with a candidacidal agent.

Caution is also adviced in patients with active organism CNS disease including epilepsy and in patients with a history of blood dyscrasias.

A mild leukopenia has been observed during its administration; however, no persistent hematologic abnormalities attributable to metronidazole have been observed in clinical studies. Total and differential leukocyte counts are recommended before and after therapy.

Regular clinical and laboratory monitoring (especially leukocyte count) are advised if administration of this medication for more than 10 days is considered as necessary.

Prescribing this medication in the absence of a proven or strongly suspected bacterial or parasitic infection or a prophylactic indication is unlikely to provide benefit to the patient and increases the risk of the development of drug-resistant bacteria and parasites.

Cases of severe bullous skin reactions such as Steven Johnson syndrome (SJS), toxic epidermal necrolysis (TEN) or acute generalised exanthematous pustulosis (AGEP) have been reported with metronidazole. If symptoms or signs of SJS, TEN or AGEP are present, this treatment must be immediately discontinued.

There is a possibility that after *Trichomonas vaginalis* has been eliminated a gonococcal infection might persist.

The elimination half-life of metronidazole remains unchanged in the presence of renal failure. The dosage of metronidazole should therefore needs no reduction. Patients with end-stage renal disease may excrete metronidazole and metabolites slowly in the urine, resulting in significant accumulation of metronidazole metabolites. Monitoring for metronidazole associated adverse events and measurement of its plasma concentration by HPLC are recommended.

In patients undergoing haemodialysis, metronidazole and metabolites are efficiently removed during an eight hour period of dialysis. Metronidazole should therefore be re-administered immediately after haemodialysis.

No routine adjustment in the dosage need to be made in patients with renal failure undergoing intermittent peritoneal dialysis (IDP) or continuous ambulatory peritoneal dialysis (CAPD).

Patients should be warned that metronidazole may darken urine.

Use of metronidazole does not obviate the need for aspirations of pus whenever indicated.

In studies on the mutagenic potential of metronidazole, the Ames test was positive while several nonbacterial tests in animals were negative. In patients with Crohn's disease, metronidazole increased the chromosome abnormalities in circulating lymphocytes. In addition, the drug has been shown to be tumorigenic and carcinogenic in rodents. Due to inadequate evidence on the mutagenicity risk in humans, the use of this medication for longer treatment than usually required should be carefully considered and carefully assessed in each case particularly in relation to the severity of the disease and the age of patient.

Metronidazole may interfere with certain chemical analysis of serum aspartate transaminase (AST), alanine transaminase (ALT), lactate dehydrogenase (LDH), triglycerides and hexokinase glucose to give abnormally low values.

Paediatrics
When used for the treatment of anaerobic infections and amoebiasis, metronidazole has not demonstrated any paediatrics-specific problems that would limit its usefulness in children.

Geriatrics
No information is available on the relationship of age to the effects of metronidazole in geriatric patients. However, elderly patients are more likely to have an age-related decrease in hepatic function, which may require an adjustment in dosage in patients receiving metronidazole. In elderly geriatric patients, monitoring for metronidazole associated adverse events is recommended.

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PREGNANCY AND LACTATION
Pregnancy
Category B2
Metronidazole should not be given in the first trimester of pregnancy as it crosses the placenta and enters foetal circulation rapidly. As its effects on human foetal organogenesis are not known, its use in pregnancy should be carefully evaluated. Although it has not been shown to be teratogenic in either human or animal studies, such possibility cannot be excluded.

Use of metronidazole for trichomoniasis in the second and third trimesters should be restricted to those in whom local palliative treatment has been inadequate to control symptoms.

Lactation
Metronidazole is present in human milk at concentrations similar to maternal serum levels, and infant serum level can be close to or comparable to infant therapeutic levels. In view of its tumorigenic and mutagenic potential, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother. Alternatively, a nursing mother may choose to pump and discard the milk for the duration of metronidazole therapy, and for 24 hours after therapy ends and feed her infant stored human milk or formula.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE
Patients should be warned about the potential for drowsiness, dizziness, confusion, hallucinations, convulsions or transient visual disorders, and advised not to drive or operate machinery if these symptoms occur.

DRUG INTERACTIONS
Alcohol
It is recommended that metronidazole not be used concurrently with or for at least 48 hours following ingestion of alcohol; accumulation of acetaldehyde by interference with the oxidation of alcohol may occur, resulting in disulfiram-like effect such as abdominal cramps, nausea, vomiting, headache or flushing; in addition, modifications in the taste of alcoholic beverages have been reported during concurrent use.

Anticoagulants, coumarin or indandione-derivative
Effects may be potentiated when these agents are used concurrently with metronidazole, because of inhibition of enzymatic metabolism of anticoagulants; periodic prothrombin time determinations may be required during therapy to determine if dosage adjustments of anti-coagulants are necessary.

Disulfiram
It is recommended that metronidazole not be used concurrently with, or for 2 weeks following disulfiram in alcoholic patients; such use may result in confusion and psychotic reactions because of combined toxicity.

Cimetidine
Hepatic metabolism of metronidazole may be decreased when metronidazole and cimetidine are used concurrently, possibly resulting in delayed elimination and increased serum metronidazole concentrations; monitoring of serum concentrations as a guide to dosage is recommended since dosage adjustment of metronidazole may be necessary during and after cimetidine therapy.

Phenobarbital
Phenobarbital may induce microsomal liver enzymes, increasing metronidazole's metabolism and resulting in a decrease in half-life plasma concentration.

Phenytoin
Metronidazole may impair the clearance of phenytoin, increasing phenytoin's plasma concentration.

Lithium
Lithium retention accompanied by evidence of possible renal damage has been reported in patients treated simultaneously with lithium and metronidazole. Lithium treatment should be tapered or withdrawn before administering metronidazole. Plasma concentrations of lithium, creatinine and electrolytes should be monitored in patients under treatment with lithium while they receive metronidazole.

5 fluorouracil
Metronidazole reduces the clearance of 5 fluorouracil and can therefore result in increased toxicity of 5 fluorouracil.

Ciclosporin
Patients receiving ciclosporin are at risk of elevated ciclosporin serum levels. Serum ciclosporin and serum creatinine should be closely monitored when coadministration is necessary.

Busulfan
Plasma levels of busulfan may be increased by metronidazole which may lead to severe busulfan toxicity. Metronidazole should not be administered concomitantly with busulfan unless the benefit outweighs the risk. If no therapeutic alternatives to metronidazole are available, and concomitant administration with busulfan is medically needed, frequent monitoring of busulfan plasma concentration should be performed and the busulfan dose should be adjusted accordingly.

DOSAGE AND ADMINISTRATION
Tablets are to be taken with or after food.

Amoebiasis, Balantidiasis, Blastocystis hominis infection
Adults: 400 to 800 mg three times daily for 5-10 days.
Children: 35 to 50 mg per kg body-weight daily in divided doses.

Giardiasis
Adults: 2 g daily as a single dose for 3 successive days.
Children: 15 mg per kg body-weight daily in divided doses.

Trichomoniasis
Adults: Single 2 g dose or a 7-day course of 200 mg three times daily or 400 mg twice daily or 2-day course of 800 mg in the morning and 1.2 g in the evening. Sexual partners should be treated concomitantly.
Children: 15 mg per kg body-weight daily in divided doses for 7 days.

Bacterial vaginosis
Adults: Single 2 g dose or 7-day course of 400 mg twice daily.

Acute necrotising ulcerative gingivitis, acute dental infections
200 mg three times daily for 3 days.

Anaerobic bacterial infections
Adults: Initial dose of 800 mg followed by 400 mg every 8 hours for 7 days.
Children: 7.5 mg per kg body-weight every 8 hours.

OVERDOSAGE

Symptoms:
Single oral doses of metronidazole up to 15 g have been reported in suicide attempts and accidental overdoses. Symptoms reported include nausea, vomiting and ataxia.

Oral metronidazole has been studied as a radiation sensitiser in the treatment of malignant tumours. Neurotoxic effects, including seizures and peripheral neuropathy, have been reported after 5 to 7 days of doses of 6 to 10.4 g every other day.

Treatment:
There is no specific antidote for metronidazole overdose, therefore, management of the patient should consist of symptomatic and supportive therapy.

ROUTE OF ADMINISTRATION
Oral

STORAGE CONDITIONS
Store below 30°C.
Keep container tightly closed.
Protect from light.

SHELF LIFE
Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION
Pharmaniaga Metronidazole tablet 200 mg
In bottle of 1000 tablets (for export only).
In blisters of 100 and 500 tablets.

Pharmaniaga Metronidazole tablet 400 mg
In bottle of 500 and 1000 tablets (for export only).
In blisters of 500 tablets.

Date of revision: 12th December 2019

PRODUCT REGISTRATION HOLDER/MANUFACTURER:
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